

### Technical Specification

For

Static 3 Phase 4 Wire 11 kV CT & PT Operated AMR Compatible DLMS Compliant  
Category C1 Tri Vector Energy Meter For Bulk Consumer

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## 1.0 SCOPE:

This specification shall cover design, engineering, manufacture, assembly, inspection, testing at manufacturers works before dispatch, supply and delivery at destination anywhere in the state of West Bengal of 0.5s accuracy class static 3 phase 4 wire CT&PT operated AMR compatible DLMS compliant Category – C Tri-Vector Energy Meter fitted with Pilfer Proof Meter Box for 11 KV Bulk Consumers for measurement of different electrical parameters listed elsewhere in the document including Active Energy(kWh), Reactive Energy (kVArh), Apparent Energy(kVAh) etc. over a Power Factor range of Zero Lag to Unity to Zero Lead. Meters shall have facility of recording tamper information & load survey of active energy, apparent energy, reactive energy, phase currents, phase voltages & other parameters in nonvolatile memory. These meters shall have communication port to interface for remote meter reading.

## 2.0 APPLICABLE STANDARDS:

Unless specified elsewhere in this specification, the performance & testing of the CT&PT operated meters of accuracy class 0.5s shall conform to the following Indian / International standards, to be read with up-to-date and latest amendments / revisions thereof as on 90 days prior to floating of tender.

|                                                     |                                                                                                                                                                                   |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IS 14697: 1999 read with latest amendments          | Specification for AC Static Transformer operated Watt Hour & VAR Hour meters, Class 0.5s                                                                                          |
| CBIP Report No. 325 & its latest amendments, if any | Specification for AC Static Electrical Energy Meters                                                                                                                              |
| IS 15959: 2011 read with latest amendments          | Data Exchange for Electricity Meter - Reading Tariff and Load Control - Companion Specification                                                                                   |
| IEC 62056(Part 21, 31, 61)                          | Electricity metering: Data exchange for meter reading, tariff and load control: Direct local data exchange, Local Area Network data exchange, Object identification system (OBIS) |
| IEC 62052-11                                        | Electricity metering equipment (AC) -General requirements, tests and test conditions -Part 11 : Metering equipment                                                                |
| IEC 62053-22                                        | Electricity metering equipment (AC) -Particular requirements - Part-22 : Static Meters for Active Energy (Class 0.5S)                                                             |
| IS-15707                                            | Specification for Testing, evaluation, installation & maintenance of AC Electricity Meters-Code of Practice                                                                       |
| IS 13410 :1992 with latest amendment                | Specification for Pilfer Proof Meter box suitable for 3 Phase Static Energy Meter                                                                                                 |
| IS 12346 :1988                                      | Specification for testing equipment for AC Static Electrical Energy Meter (latest amendment)                                                                                      |

## 3.0 CLIMATIC CONDITION:

The meter shall be suitably designed for satisfactory operation under the hot and hazardous tropical climate conditions and shall be dust and vermin proof. All the parts and surface, which are subject to corrosion, shall either be made of such material or shall be provided with such protective finish, which provided suitable protection to them from any injurious effect of excessive humidity.

The meter shall be suitable for satisfactory continuous operation under the following tropical conditions.

- 3.1 Maximum Ambient Air Temperature in shade: 55° C
- 3.2 Minimum Ambient Air Temperature: (-)10° C
- 3.3 Maximum Relative Humidity: 95%(non-condensing)
- 3.4 Minimum Relative Humidity: 10%
- 3.5 Height above mean sea level: Up to 3000 meters
- 3.6 Average number of months for Tropical Monsoon per Annum: 5
- 3.7 Annual Rainfall: 100 mm to 1500 mm
- 3.8 Maximum Wind Pressure: 150 Kg/Sqm

## 4.0 SUPPLY SYSTEM:

|        |                |
|--------|----------------|
| System | 3 Phase 4 Wire |
|--------|----------------|

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|                 |                                                                                                      |                                                     |
|-----------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Rated Voltage   | Reference Voltage (Vref)<br>PT Ratio: 11kV/ $\sqrt{3}$ / 110 V/ $\sqrt{3}$                           | 63.5 V – Phase to Neutral<br>110 V – Phase to Phase |
| Rated Current   | Basic Current (I <sub>b</sub> ) : 5A, Maximum Current (I <sub>max</sub> ) : 10A<br>CT Ratio : 50/5 A |                                                     |
| Rated Frequency | 50 Hz                                                                                                |                                                     |

#### 5.0 POWER FACTOR RANGE:

The meter shall be lag only suitable for full power factor range from zero (lagging) through unity to zero (leading).

#### 6.0 POWER SUPPLY VARIATION:

The meter shall be suitable for working with following supply system variations.

|                              |                                                 |
|------------------------------|-------------------------------------------------|
| System                       | 3 Phase 4 Wire                                  |
| Specified range of operation | 70% to 120% of reference Voltage i.e. 63.5 Volt |
| Frequency                    | 50Hz $\pm$ 5%                                   |

#### 7.0 ACCURACY:

- 7.1 Class of accuracy of the meter shall be 0.5s
- 7.2 Maximum error limit at 1% I<sub>b</sub>, UPF shall not exceed  $\pm$ 1%
- 7.3 There shall be no drift in accuracy, at least for a period of ten years from the date of supply. In case any drift is noticed within the specified guarantee period, the successful bidder shall have to replace the meter with a new one free of cost.

#### 8.0 POWER CONSUMPTION:

- 8.1 Voltage Circuit: The active and apparent power consumption in the voltage circuit/phase including the power supply of meter along with split unit at reference voltage, reference temperature and reference frequency shall be as per IS 14697.
- 8.2 Current Circuit: The apparent power taken by current circuit/phase at basic current, reference frequency and reference temperature shall be as per IS 14697.

#### 9.0 STARTING CURRENT & RUNNING AT NO LOAD:

- 9.1 The meter shall start registering energy at 0.1% of basic current at unity power factor and first pulse must appear within 10 minutes.
- 9.2 Running at no load: When 70% & 120% voltage is applied and no current flows in the current circuit, the test output of the meter shall not produce more than one pulse.

#### 10.0 MAXIMUM CONTINUOUS CURRENT:

The maximum continuous current in meters shall be the current at which the meter purports to meet the accuracy requirement of the specification. The same is indicated in table in Supply System Clause of this specification.

#### 11.0 GENERAL & CONSTRUCTIONAL REQUIREMENTS:

- 11.1 The case, winding, voltage circuit, sealing arrangements, registers, terminal block, terminal cover & name plate etc. shall be in accordance with the relevant standards. The meter shall be compact & reliable in design, easy to transport & immune to vibration & shock involved in the transportation & handling. The construction of the meter shall ensure consistence performance under all conditions especially during heavy rains / very hot weathers. The insulating materials used in the meter shall be non-hygroscopic, non-ageing & have tested quality. Under normal condition of use winding & insulation shall not reach a temperature which might adversely after the operation of the meters.
- 11.2 The meter shall be sealed in such a way that the internal parts of the meter becomes inaccessible and attempts to open the meter shall result in viable damage to the meter cover i.e. break to open type. This is to be achieved by using continuous Ultrasonic welding on all the four sides of the Meter base and cover or any other technology which is either equally or more efficacious.
- 11.3 The meter shall comply with latest technology such as Microcircuit or Application Specific Integrated Circuit (ASIC)



to ensure reliable performance. The mounting of the components on the PCB shall compulsorily be of Surface Mounted Technology (SMT) type. Power supply component shall be of PTH type. The electronic components used in the meter shall be of high quality and there shall be no drift in the accuracy of the meter for at least ten years. The circuitry of the meter shall be compatible with 16 Bit (or better) ASIC with compatible processor and meter shall be based on Digital measuring and sampling technique.

- 11.4 The meter shall be housed in a safe, high grade, unbreakable, fire resistant, UV stabilized, virgin Polycarbonate casing of projection mounting type. The meter cover shall be transparent / translucent. But the viewing portion shall be transparent for easy reading of displayed parameters and observation of operation indicators. The meter base may not be transparent but it shall not be black in colour. The meter casing shall not change in shape, colour, size and dimensions when subjected to 72 hrs on UV test as per ASTM D 53. It shall withstand 650 °C glow wire test and heat deflection test as per ISO 75 or as per IEC 60068 -2-5.
- 11.5 In addition to the above, the meter cover shall be sealable to the meter base with at least 2 nos. bar coded seals bearing the identification marks of the Manufacturer. Suitable arrangement shall be made for fitting/fixing of utility seal at two sides of meter terminal cover in such a manner that any access to the terminal cannot be made possible without removing the seal. There shall also be provision for sealing at the optical port.
- 11.6 The polycarbonate material of only the following manufacturers shall be used:
  - 11.6.1 G.E. PLASTIC: LEXAN 943A or equivalent like 943, 123R, 143 for meter cover & terminal cover / LEXAN 503R or equivalent like 500, 143R, 500R for meter base and terminal block.
  - 11.6.2 BAYER : Grade corresponding to above
  - 11.6.3 DOW Chemical: --do--
  - 11.6.4 MITSUBISHI: --do--
  - 11.6.5 TEJIN : --do--
  - 11.6.6 DUPONT : --do--

#### 12. METER CASE AND COVER:

- 12.1 In case, ultrasonic welding using plate / strip is used, the material of plate / strip shall be same as that of cover and base and the strip. The manufacturer's logo shall be embossed on the strip / plate. The material of the meter body (case and cover) shall be of Engineering Plastic.
- 12.2 The meter cover shall be fixed to the meter base (case) with Unidirectional Screws, so that the same cannot be opened by use of screwdrivers. These unidirectional screws shall be covered with transparent caps (not required for screw less design), ultrasonically welded with the meter body and the screw covers shall be embedded in the meter body in a groove. The meter shall withstand external magnetic influence as per latest amendments of CBIP Technical Report No.325 including 0.2T AC Magnet, 0.5T Permanent magnet.

#### 13. TERMINAL BLOCK AND COVER:

- 13.1 The terminals may be grouped in a terminal block having adequate insulating properties and mechanical strength. The terminal block shall be made from best quality non-hygroscopic, flame retardant material (capable of passing the flammability tests) with nickel plated brass inserts / alloy inserts for connecting terminals. It shall be rigidly fixed to the base of the meter so that it cannot be separated from the meter base without breaking either the meter base or the terminal block and this fixing arrangement shall be in parallel to the meter base in such a way that it cannot be viewed or approached from any part of the meter without breaking the meter.
- 13.2 The terminals in the terminal block shall be of adequate length in order to have proper grip of conductor. The screws shall have thread size not less than M4 and head having 6 mm. Diameters. The screws shall not have pointed ends at the end of threads. All terminals and connecting screws and washers shall be of tinned / nickel plated brass material. The terminal shall withstand glow wire test at 960 ± 15°C and the terminal shall withstand at least 135°C as per IS.
- 13.3 The internal diameter of terminal hole shall be minimum 5.5 mm and center to center distance is 13 mm. The holes in the insulating material shall be of sufficient size to accommodate the insulation of conductor also.
- 13.4 The terminal cover shall be transparent re-enforced Polycarbonate, Engineering Plastic with minimum thickness 2.0 mm and the terminal cover shall be of extended type completely covering the terminal block and fixing holes. The space inside the terminal cover shall be sufficient to accommodate adequate length of external cables.
- 13.5 The terminals and all connecting screws will be of suitable material capable of withstanding a current of 150% of



I<sub>max</sub> for two hours, continuously and the meter shall be capable of providing phase to neutral protection up to 433 V for 1(one) hours.

#### **14.0 MARKING OF THE METER:**

The marking on the meter shall be in accordance with relevant clauses of IS 14697. The basic marking on the meter nameplate shall be as follows. All other markings as per IS shall also be there.

- 14.1 Manufacturer's name & trade mark
- 14.2 Place of manufacture
- 14.3 Month & Year of manufacture
- 14.4 Type Designation
- 14.5 No. of phases & wires
- 14.6 Serial number of meter (Size not less than 5mm)
- 14.7 Power Factor Range
- 14.8 Reference Voltage
- 14.9 PT Ratio
- 14.10 CT Ratio
- 14.11 Rated Secondary Current
- 14.12 Rated Maximum Current
- 14.13 Operating Frequency
- 14.14 Principal unit(s) of measurement
- 14.15 Meter Constant (imp/kwh)
- 14.16 Class index of meter
- 14.17 "Property of WBSEDCL"
- 14.18 Purchase Order No. & Date
- 14.19 Guarantee (Guaranteed for a period of 5 ½ years from the date of delivery)
- 14.20 BIS marking
- 14.21 Barcode for meter serial no. in alpha numeric form, date of manufacture, current rating of the meter and PO reference, readable by single layer barcode reader.
- 14.22 The reference temperature if different from 27°C.
- 14.23 The sign of Double Square for insulating encased meters
- 14.24 Firmware Version

#### **15.0 CONNECTION DIAGRAM AND TERMINAL MARKING:**

Every meter shall be indelibly marked with a diagram of connection. For this poly-phase meter, this diagram shall also show the phase sequence for which the meter is intended. It is permissible to indicate the connection diagram by an identification figure in accordance with relevant standards. The marking of meter terminals shall appear on this diagram.

#### **16.0 DISPLAY OF MEASURED VALUES:**

- 16.1 The meter shall have alphanumeric display with at least 7 full digit with LCD backlit display, having minimum character height of 10 mm. The data shall be stored in nonvolatile memory (NVM). The NVM shall retain data for a period of not less than 10 years under unpowered condition. Battery backup memory will not be considered as NVM.
- 16.2 It shall be possible to easily identify the single or multiple displayed parameters through symbols / legend on the meter display itself or through display annunciation which shall be self explanatory and symmetric.
- 16.3 Serial number of the meter shall be displayed on the display plate. It shall also be programmed into meter memory for identification through communication port(Optical / RS232 /RS11) for CMRI / laptop / meter reading printout.

- 16.4 Visibility of display in poor light conditions is an important criterion. STN or TN or any better type of advanced LCD to be used. Proper legends for the displayed parameters to be provided (Factory programmable). Back lit provided for clear visibility shall be uniform throughout all part of the LCD.
- 16.5 The meters shall have auto-display mode for pre-selected parameters. The push-button mode of display shall display all parameters and it shall have priority over auto-display mode.
- 16.6 The meter shall give clear message on display to indicate that the meter has experienced tampers and the nature of tamper with date and time of first occurrence, last occurrence and last restoration, if the last tamper status is not restored, then meter will indicate first occurrence, last restoration and last occurrence.
- 16.7 Connection check, phase sequence and self diagnostic shall give clear message on display. The meter shall have a test output (blinking LED) accessible from the front and be capable of being monitored with suitable testing equipment. The operation indicator must be visible from the front.
- 16.8 Test output device shall be provided in the form of one common LED for active and reactive energy with the provision of selecting the parameter being tested (separate LED may also be used with proper separation).

#### 17.0 DISPLAY SEQUENCE:

The meter shall display the required parameters in two different modes as follows. Display sequence for both auto and Push button must be maintained, no interchange in sequence or display parameter will be accepted. All the display shall have proper legend to identify the same.

17.1 AUTO DISPLAY MODE: The following parameters shall be displayed in auto mode in following sequence

- 17.1.1 LCD check (All segment check on display)
- 17.1.2 Connection Check /Phase Sequence (Voltage & Current)
- 17.1.3 Meter Serial Number BIS
- 17.1.4 Real Day/ Date (dd:mm:yy)
- 17.1.5 Real Time (hh:mm:ss)
- 17.1.6 Rising Demand with elapsed time
- 17.1.7 R-Phase to Neutral Voltage
- 17.1.8 Y-Phase to Neutral Voltage
- 17.1.9 B-Phase to Neutral Voltage
- 17.1.10 R-Phase Line Current
- 17.1.11 Y-Phase Line Current
- 17.1.12 B-Phase Line Current
- 17.1.13 Instantaneous Average PF
- 17.1.14 Instantaneous Active Demand
- 17.1.15 Instantaneous Apparent Demand
- 17.1.16 Total Active Forward Energy
- 17.1.17 TOD-wise Total Active Forward Energy
- 17.1.18 Total Apparent Forward Energy
- 17.1.19 TOD-wise Total Apparent Forward Energy
- 17.1.20 Active MD with Date & Time [0-24 hrs.]
- 17.1.21 Average PF Associated with Active MD Date & Time Slot [0-24 hrs.]
- 17.1.22 TOD-wise Active MD with Date & Time
- 17.1.23 Average PF Associated with TOD-wise Active MD Date & Time Slot
- 17.1.24 Apparent MD with Date & Time [0-24 hrs.]
- 17.1.25 Average PF Associated with Apparent MD Date & Time slot [0-24 hrs.]
- 17.1.26 TOD-wise Apparent MD with Date & Time
- 17.1.27 Average PF Associated with TOD-wise Apparent MD Date & Time Slot

17.2 PUSH BUTTON MODE: (In addition to the auto display mode parameters the following parameters shall be displayed on pressing the Push Button)

- 17.2.1 History 1 Total Active Forward Energy

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- 17.2.2 History 1 TOD-wise Active Forward Energy
- 17.2.3 History 1 Total Apparent Forward Energy
- 17.2.4 History 1 TOD-wise Apparent Forward Energy
- 17.2.5 History 1 Active MD with Date & Time [0-24 hrs.]
- 17.2.6 History 1 Average PF Associated with Active MD Date & Time Slot [0-24 hrs.]
- 17.2.7 History 1 TOD-wise Active MD with Date & Time
- 17.2.8 History 1 Average PF Associated with TOD-wise Active MD Date & Time Slot
- 17.2.9 History 1 Apparent MD with Date & Time [0-24 hrs]
- 17.2.10 History 1 Average PF Associated with Apparent MD Date & Time Slot [0-24 hrs.]
- 17.2.11 History 1 TOD-wise Apparent MD with Date & Time
- 17.2.12 History 1 TOD wise Average PF Associated with Apparent MD Date & Time Slot
- 17.2.13 History 1 Average PF
- 17.2.14 History 1 Average LF
- 17.2.15 History 1 Power Off Event Duration
- 17.2.16 History 1 Power Off Event Count
- 17.2.17 MD Reset Time(hh:mm:ss) & Date(dd:mm:yy)
- 17.2.18 Cumulative Apparent MD with Date & Time
- 17.2.19 MD Reset or Bill Count
- 17.2.20 Cumulative Tamper Count
- 17.2.21 Cumulative Programming Count
- 17.2.22 Cumulative Power Off Duration (hhhh:mm)/(dddd:hh:mm)
- 17.2.23 Cumulative Power Off Count
- 17.2.24 R-Phase PF
- 17.2.25 Y-Phase PF
- 17.2.26 B-Phase PF
- 17.2.27 Supply Frequency
- 17.2.28 Average PF [Current Month]
- 17.2.29 Present Power Related Tamper Status
- 17.2.30 Present Current Related Tamper Status
- 17.2.31 Present Voltage Related Tamper Status
- 17.2.32 Present Other Status
- 17.2.33 First Occurred Tamper EventID
- 17.2.34 Date of Occurrence of First Tamper Event
- 17.2.35 Time of Occurrence of First Tamper Event
- 17.2.36 Lastly Occurred Tamper Event
- 17.2.37 Date of Occurrence of Last Tamper Event
- 17.2.38 Time of Occurrence of Last Tamper Event
- 17.2.39 Lastly Restored Tamper Event
- 17.2.40 Date of Restoration of Lastly Restored Tamper Event
- 17.2.41 Time of Restoration of Lastly Restored Tamper Event
- 17.2.42 High Resolution Cumulative Active Forward Energy
- 17.2.43 High Resolution Cumulative Reactive Lag Energy
- 17.2.44 High Resolution Cumulative Reactive Lead Energy
- 17.2.45 High Resolution Cumulative Apparent Energy
- 17.2.46 Date of Occurrence of Front Cover Open Tamper Event
- 17.2.47 Time of Occurrence of Front Cover Open Tamper Event
- 17.2.48 Self-Diagnostic Feature

### 17.3 MANUAL BATTERY MODE/SUB ACTIVE MODE:

- 17.3.1 Meter Serial Number BIS

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- 20.1 Electromagnetic field
- 20.2 External magnetic field
- 20.3 Radio frequency interference
- 20.4 Vibration
- 20.5 Harmonic Distortion
- 20.6 Voltage variation (70% - 120% of Vref.)
- 20.7 Frequency variation (+/-) 5% of 50 Hz

#### **21.0 RESETTING OF MAXIMUM DEMAND:**

- 21.1 The meter shall be capable of recording the Apparent MD with integration period of 15 minutes,
- 21.2 The meter shall be capable of recording the Actual Average Power factor of the time block corresponding to the period of recording the maximum demand
- 21.3 MD reset shall be through each of the three means:
  - 21.3.1 Automatic resetting at preset date & time (at present it will be at 00.00 hrs of the first day of the month)
  - 21.3.2 Manually i.e., by Push Button
  - 21.3.3 Through authenticated command from CMRI or through Remote Communication.
- 21.4 The means by which the reset has been done shall be made available to downloaded data. Facility to invoke any of the above through authenticated MRI command shall be provided at BCS.
- 21.5 MD reset button shall have proper sealing arrangement.
- 21.6 There shall be separate Push button for scrolling display (up and down) and MD reset. If only two Push buttons are used minimum 60 sec pressing is required for MD reset

#### **22.0 LOAD SURVEY:**

The meter shall be capable of recording load survey for the following parameters for a period of minimum 60 POWER ON days - subject to availability of all parameters listed below with 15 minutes integration period.

- 22.1 Active Energy in kWh
- 22.2 Reactive Energy (lag & lead) in rkVAh
- 22.3 Apparent Energy in kVAh
- 22.4 Demand in kVA and kW
- 22.5 Phase-wise Current
- 22.6 Phase-wise Voltage
- 22.7 Average Power Factor

Real time & date shall be shown in all the load survey graphs and tables. All the load survey data shall be available in numerical as well as graphical format. It shall be possible to view this data in daily, weekly and monthly formats. The load survey graph shall show value.

#### **23.0 NON VOLATILE MEMORY & REAL TIME CLOCK BATTERY:**

The Non Volatile Memory (NVM) shall not require any additional battery backup to retain the data in case of power failure at least for up to 10 years and the data storage shall be independent of battery backup unit. The life of the Real Time Clock (RTC) battery in circuit condition shall be minimum 10 years in case of power failure. It shall be possible to transfer this data to base computer software through MRI/Laptop or RMR. The data so obtained shall be displayed in both graphical & numeric form in the BCS. The BCS with all details is to be provided by the supplier at no extra cost.

#### **24.0 TIME OF DAY FACILITIES:**

The meter shall have facility to record Active & Apparent Energy and MD in at least 8 time zones. Out of 8 TOD zone only 3 TOD zones will remain active and other the other TOD zones will remain dormant for future provision. The time zones shall be user programmable through authenticated CMRI/Laptop/RMR command. Necessary software for the same

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is to be provided by the successful bidder.

Default settings for the TOD timings shall be as follows:

- 24.1 TOD 1: 06:00 Hrs to 17:00 Hrs.
- 24.2 TOD 2: 17:00 Hrs to 23:00 Hrs.
- 24.3 TOD 3: 23:00 Hrs to 06:00 Hrs.

#### **25.0 METER READING DURING POWER OFF:**

It shall be possible to read the meter-display visually and with MRI/Lap top in absence of input voltages with the help of internal battery for display in power off condition or external battery pack/ PPU. If internal battery is used then it shall be capable of supplying the required power at least two years under meter un-powered condition and subsequent powering up of the battery shall guarantee it's life till the expiry of guarantee period. In case of external battery the arrangements shall be such that hands free operation is possible. In case of external battery 10 years guarantee must be given for external battery/power pack. Separate battery shall be used for this purpose (Not RTC or processor battery). In case of Lithium battery rating shall be more than 500mAh. Any downloading problem due to battery shall be treated as defective within Guarantee Period.

#### **26.0 SELF DIAGNOSTIC FEATURES:**

The meter shall be capable of performing complete self-diagnostic check to monitor the circuits for any malfunctioning to ensure integrity of data memory location all the time. If possible, the details of malfunctioning shall be recorded in the meter memory. The bidder shall furnish the details of self-diagnostic capability feature, viz Memory status (NVM) and Battery status, RTC Status etc. and it shall be in display.

#### **27.0 COMMUNICATION CAPABILITY:**

- 27.1 The meter shall have a galvanically Isolated optical communication port as per IEC 1107/ANSI/PACT so that it can be easily connected to a hand-held common meter reading instrument (CMRI) for data transfer. The data downloading time shall be less than 5 minutes. The optical port shall be provided with proper sealing arrangement so that the optical cover shall not be opened without breaking the seal. The stored data in the meter shall be available through CMRI even when the display of the meter is not available.
- 27.2 A separate suitable serial port (RS-232/RJ11) capable of being hooked into a remote metering device such as modem, shall be provided inside the terminal cover to enable Automatic Meter Reading in such a way that the same cannot be accessed without opening the Terminal Cover and Seal.
- 27.3 Both the ports will have equal priority in terms of data accessing. Data transmission rate shall be 9600 bps for both ports.
- 27.4 It shall not be possible to alter date in the meter bypassing commands from the CMRI or Laptop. Correction of RTC time, change of TOD timing, etc. shall be possible through CMRI/Laptop using authenticated command sets from BCS after scheduling for particular meter serial no. Billing parameters shall be factory programmable. No change shall be possible using CMRI only, i.e. the control has to be with the BCS. Any such change shall be recorded as a transaction and to be recorded in meter memory and also shall be available at subsequent BCS data with date & time stamping.

#### **28.0 BASE COMPUTER SYSTEM & SOFTWARE REQUIREMENTS:**

- 28.1 The Common Meter Reading Instrument (CMRI) shall be loaded with user-friendly MS-DOS 5.0( or its higher version) compatible software for reading and/or downloading meter data.
- 28.2 Windows 7.0 (or its higher version) based Base Computer Software (BCS) shall be provided for receiving data from CMRI and downloading instructions from BCS to CMRI.
- 28.3 The data stored in the meter memory including defrauded energy shall be available on the BCS housed on a PC or Laptop either directly from meter or through CMRI.
- 28.4 Only one BCS shall be provided for downloading meter data and issue of authenticated commands to meter through CMRI/PC/Laptop.



- 28.5 Before performing any transaction through authenticated command from CMRI or Laptop/PC, the software shall be taking full meter data as backup and the same shall be available at BCS end.
- 28.6 The BCS shall have facility to convert meter data into user definable ASCII file format so that it can be integrated with the billing system or any other third party software. The user shall have the flexibility to select the parameters to be converted into ASCII file. A wizard shall be available whereby user can select file format, data to be exported, field width of the selected parameters in a user friendly manner.
- 28.7 All the data available in the meter ( including parameters like energy, demand, voltage, current and tamper events etc.) with date and time stamp, new TOD time zones and historical data shall be available in BCS after downloading.
- 28.8 The bidder shall supply the necessary CMRI software during sample meter testing.
- 28.9 The bidder has to supply the meter reading protocol and API free of cost. The bidder shall indicate the relevant standard to which the protocol is compliant.
- 28.10 Transfer of data from the meter directly or through CMRI to PC/Laptop shall have to be trouble free and user friendly.
- 28.11 Any change or up gradation of CMRI software or BCS in future, required for any reason, has to be made by the supplier at his own cost and subsequently provided to the purchaser.
- 28.12 The same software shall be capable of preparing CMRI to read the meter information or to reconfigure the meter for change of TOD timings and / or time setting of the meter etc.
- 28.13 The exhaustive online help shall be available with the software so that user can use all the features of the software by just reading the help contents.
- 28.14 In BCS, history data containing total & TOD wise cumulative, instantaneous, average, maximum values of all vital electrical parameters mentioned anywhere in this technical specification (like energy, demand, maximum demand, power factor, load factor, voltage, current , power failure duration, power failure event count etc. ) & all event records along with snapshots , all with date & time stamp for at least last twelve must be made available.
- 28.15 Midnight fundamental & total energy parameters, both cumulative and TOD wise, need be made available at BCS end.
- 28.16 All load survey data need be made available at BCS end in numerical as well as graphical format with date & time stamp. The load survey graphs shall show values.
- 28.17 It shall be also possible to view these data in daily, weekly and monthly formats through BCS.

#### **29.0 SUPPLY OF HHU/ CMRI, COMMUNICATION CORD (HHU TO PC AND METER TO LAPTOP / PC):**

For every 100 nos. of meters and part thereof one CMRI with not less than 32 MB memory (DOS based). Meter to PC USB type communication cord with Moxa converter for each type should be supplied free of cost with each CMRI. The guarantee period of HHU/CMRI and Power Pack shall be 66 months from the date of supply.

#### **30.0 ANTI TAMPER FEATURES:**

The meter shall have the following anti-tamper features:

- 30.1 The meter shall be capable of recording energy correctly even if the input and output terminals are interchanged.
- 30.2 The meter shall work correctly irrespective of phase sequence of supply. There must be an indication of reversal of phase sequence in display as well as in downloaded data.
- 30.3 The meter shall work correctly even in absence of Neutral.
- 30.4 The meter shall record energy within maximum error of + 4% on injection of DC, pulsating DC (7-10 Hz), chopped AC in Neutral along with logging of ND tamper. In case tamper event is not logged, i.e. meter is immune to neutral disturbance, accuracy of the meter must not get affected. Maximum chopping for AC injection will be 25% to 30% at peak end.
- 30.5 The energy registration shall not be affected more than + 4% if high frequency (55Hz to 100Hz) or low frequency (45Hz to 30 Hz) AC signal w.r.to earth is applied to the meter neutral.
- 30.6 The meter shall be immune to Electro Static Discharge or sparks of 35 kV (approx) induced by using frequency-generating devices having very high output voltage.

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- 30.7 Tests in this respect will be conducted by using commonly available devices and during spark discharge test, spark will be applied directly at all vulnerable points of the meter for a period of 10 minutes (at an interval of 1 minute (approx) between two consecutive strokes) and meter shall maintain accuracy after the test under this condition. Accuracy will be checked during and after application of spark discharge Test. Meter shall record correctly within the specified limits of errors. Beyond 35 KV the meter shall record tamper if not immune.
- 30.8 The meter shall be capable of recording occurrence and restoration with date and time in respect to the following tamper events:
- 30.8.1 Power failure (Tamper count not to be increased)
  - 30.8.2 Invalid Voltage
  - 30.8.3 Missing Potential (phase wise)
  - 30.8.4 High Voltage
  - 30.8.5 Low Voltage
  - 30.8.6 Voltage Unbalance
  - 30.8.7 CT Open
  - 30.8.8 CT Bypass/ CT Short
  - 30.8.9 CT Unbalance
  - 30.8.10 CT Reversal
  - 30.8.11 Over Current
  - 30.8.12 Low Power Factor
  - 30.8.13 Neutral Disturbances (If it is logged)
  - 30.8.14 Magnetic Disturbances
  - 30.8.15 Invalid Phase Association
  - 30.8.16 Cover Open (no restoration)
  - 30.8.17 35 kV ESD
  - 30.8.18 Jammer
- 30.9 Threshold values of all above occurrence and restoration are attached elsewhere in this document. Snapshot values of Phase Voltage, Phase Current & Phase wise Power Factor, Active Energy value during occurrence & restoration to be provided in all the above mentioned tamper conditions in BCS with date and time. (In Event logging Snapshots shall be considered when the actual phenomenon occurred). The logging time for recording occurrence and restoration of all tamper events except Magnetic & Neutral Disturbance tamper, shall be 5 min. Magnetic tamper shall appear instantaneously, Neutral Disturbance within 3 min.
- 30.10 All authenticated commands shall be Base Computer Software (BCS) controlled. All transactions with meter shall be date and time logged, in the downloaded data (Last 12 month's transactions).
- 30.11 Properly designed meter tamper logic shall be provided and clearly explained in the bid. More than one tamper (CT related/ PT related/ others) shall not be logged at a time.
- 30.12 A minimum of 400 events (one event means either occurrence or restoration) of all types of tamper with date & time stamping shall be available in meter memory compartment wise.
- 30.13 The logging of events will be on FIFO basis. The events will be divided into three compartments like CT related (148 Events), PT related (88 Events), Power Failure related (100) and others (64 Events).
- 30.14 Meter shall have a continuous and clear indication in its display if top cover is removed / opened and shall be logged in BCS.

### 31.0 MANUFACTURING ACTIVITIES:

- 31.1 All the materials, electronics and power components, ICs used in the manufacture of the meter shall be of highest quality and reputed make to ensure higher reliability, longer life and sustained accuracy. The manufacturer shall use Application Specific Integrated Circuit (ASIC) or Micro controller for metering functions.
- 31.2 The electronic components shall be mounted on the printed circuit board using latest Surface Mounted Technology (SMT) except power components by deploying automatic SMT pick and place machine and re flow solder process. The electronic components used in the meter shall be of high quality and there shall be no drift in the accuracy of the meter at least up to 10 years.



- 31.3 Further, the Bidder shall own or have assured access (through hire, lease or sub-contract) of the mentioned facilities. The PCB material shall be of glass epoxy FR-4 grade conforming to relevant standards.
- 31.4 All insulating materials used in the construction of meters shall be non-hygroscopic, non-ageing and tested quality. All parts that likely to develop corrosion shall be effectively protected against corrosion by providing suitable protective coating. Quality shall be ensured at different stages. At PCB manufacturing stage, each board shall be subjected to bare board testing. At insertion stage, all components shall undergo testing for conforming to design parameters and orientation. Complete assembled and soldered PCB shall undergo functional testing using test equipment (testing jig).
- 31.5 Prior to final testing and calibration, all meters shall be subjected to accelerated ageing test to eliminate infant mortality, i.e., meters are to be kept in ovens for 72 hours at 55 deg Centigrade temperature & atmospheric humid condition. After 72 hours meters shall work correctly. Facilities / arrangement for conducting ageing test shall be available with the manufacturer.
- 31.6 The calibration of meters shall be done in-house.
- 31.7 The bidder shall submit the list of components used in the meter along with the offer. A detailed list of bought-out items, which are used in the manufacture of the meter, shall be furnished indicating the name of firms from whom these items are procured. The bidder shall also give the details of quality assurance procedures followed by him in respect of the bought-out items.
- 31.8 The details of testing facilities available for conducting the routine and acceptance tests and other special tests on the meter shall be furnished with the bid. The facility available if any for conducting type test may also be furnished.

### 32.0 **GUARANTEE:**

- 32.1 The Meters and Pilfer Proof Meter Boxes shall be guaranteed for a period of 5½ years from the date of supply against faulty design, poor materials, bad workmanship.
- 32.2 The supplied accessories, like HHU and Power Pack shall also be guaranteed for a period of 5½ years from the date of supply.
- 32.3 Bank Guarantee shall remain valid for 7½ years with a claim period of 6 months.
- 32.4 The Meters and /or Pilfer Proof Meter Boxes found defective within the above guarantee period shall have to be replaced by the supplier free of cost within one month on receipt of intimation.
- 32.5 If the defective Meters and / or Pilfer Proof Meter Boxes are not replaced within the above specified period, WBSEDCL will recover twice the cost of Meters and/or Pilfer Proof Meter Boxes from the supplier.
- 32.6 Name plate of the meter is to be marked with "Guarantee of the Meter": 5½ years from the date of supply.

### 33.0 **TYPE TESTING OF METER:**

The offered meters shall be type tested at any NABL accredited laboratory in accordance with IS 14697:1999 and CBIP Report 325 with their latest amendments. The type test report shall not be more than 5(five) years old. A copy of the Type Test report shall have to be enclosed with the offer. If there is any modification in the design / parameters of the specifications or use of constituent materials in the offered meter submitted with the offer, from the meter which was submitted type tested, that may affect the characteristics as well as parameters of the meter, revised type test certificate as per the design, parameters and constituent material used in the offered meter, shall have to be submitted failing which the offer may be liable to be rejected. Type Test Certificate from any NABL accredited Lab shall only be considered. Type test certificate shall contain the following information clearly:

- i. Type of display or LCD
- ii. Class of accuracy
- iii. Meter constant
- iv. Type of meter

### 34.0 **ACCEPTANCE TEST:**

The acceptance tests as stipulated in C9BIP / IS (with latest amendments) shall be carried out by the supplier in presence of purchaser's representative. In case of failure of Meters as specified in Recommended Sampling Plan of IS-14697:1999, the entire lot will be treated as rejected. Also the following additional tests are to be carried out on one meter randomly

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selected from each lot offered for inspection / acceptance testing. In case of failure of any single meter the entire lot will be rejected.

- 34.1 Magnetic induction of external origin (AC & DC)
- 34.2 Tamper & Fraud protection, as per relevant Clause of this specification.
- 34.3 Test of endurance up to 150% of I<sub>max</sub>, for two hours, followed by verification of limits of error
- 34.4 Verification of internal components
- 34.5 Dry Heat Test under Test of Climatic Influences in IS 14697: 1999 of one meter from the offered lot is to be arranged by the supplier at any NABL accredited laboratory, at his cost.

### 35.0 ROUTINE TEST:

Each and every meter of the offered lot shall undergo the routine tests as well as functional tests as per IS: 14697:1999, CBIP Report 325 and after sealing the meters, the manufacturers will have to submit the routine test report of all the meters as well as a statement showing seal Sl. Nos. against each meter Sl. No. of offered lot in soft copy (MS WORD or EXCEL format), to the Chief Engineer( Procurement and Contract) and the Chief Engineer( DTD), along with offer letter for acceptance test.

### 36.0 TEST FACILITIES:

- 36.1 The tests of equipment / instrument shall be carried out as per relevant Standards and test certificates shall be furnished for scrutiny. The Bidder shall indicate the details of the equipment available with him for carrying out the various tests as per relevant Standards. The bidder shall indicate the sources of all equipment / instruments.
- 36.2 The standard meters used for conducting tests shall be calibrated periodically at any NABL Accredited Test Laboratories and test certificates shall be available at Works for verification by purchasers' representative.
- 36.3 The manufacturer shall have at least the following testing facilities to ensure accurate calibration:-
- 36.4 AC high voltage test
- 36.5 Insulation test
- 36.6 Test of No Load condition
- 36.7 Test of Starting Current
- 36.8 Test on Limits of error ( Automatic Testing facility with ICT)
- 36.9 Power loss in voltage and current circuit
- 36.10 Test of Repeatability of error
- 36.11 Test of meter constant
- 36.12 Test of magnetic influence (As per CBIP 325 & Permanent Magnet)

### 37.0 INSPECTION:

- 37.1 The purchaser may carry out the inspection at any stage of manufacture. The manufacturer shall grant free access to the purchaser's representative at a reasonable time when the work is in progress. Inspection and acceptance of any equipment under this specification by the purchaser shall not relieve the supplier of his obligation of furnishing the equipment in accordance with the specification and shall not prevent subsequent rejection if the equipment is found to be defective. All acceptance tests and inspection shall be made at the place of manufacturer unless otherwise especially agreed upon by the Bidder and purchaser at the time of purchase. The Bidder shall provide all reasonable facilities without charge to the inspector, to satisfy him that the equipment is being furnished in accordance with this specification.
- 37.2 The supplier shall keep the purchaser informed in advance, about the manufacturing program for each lot so that arrangement can be made for inspection. The purchaser reserves the right to insist for witnessing the acceptance / routine testing of the bought out items. The supplier shall give 15 days for local supply / 30 days in case of foreign supply advance intimation to enable the purchaser to depute his representative for witnessing the acceptance and routine tests.
- 37.3 The purchaser reserves the right to get type test any meter, for meter casing etc. from any of the offered lots,



reserve at any destination stores.

**38.0 SUBMISSION OF SAMPLE METER:**

- 38.1 The bidder will submit his sample Meters in sealed casing / cartoon along with relevant Meter documents (As per Annexure-IV), on any working day, from 11.00 A.M. to 04.00 P.M. on weeks days within the specified period of submission latest by 01.00 P.M. on the last day of submission of bid to the Office of the Chief Engineer (DTD), Abhikshan, Sec-V, Salt Lake, Kolkata-91. The bidder will be given a receipt, jointly signed by the bidder and DTD officials, mentioning the samples and papers submitted by the bidder as per check list.
- 38.2 While submitting the samples and required documents as per Annexure-IV, the bidder shall submit three number of sealed meters as per the specifications stated herein before, two number of meters welded ultrasonically or by any other accepted technique and one number of meter, not welded, for physical verification of components and another dummy prototype meter case (for checking quality of welding) and one P.P. Box.
- 38.3 The date of testing of sample meters will be intimated to the bidders by CE (DTD) and during such test other bidders will also be allowed to witness the testing. Sample submission and Test procedure may be changed due to emergency requirement.
- 38.4 On the date of testing of sample meters of a particular bidder, he shall come prepared with the following:
- 38.4.1 BCS (as per specification)
  - 38.4.2 CMRI compatible with BCS and loaded with CMRI software
  - 38.4.3 Laptop compatible with BCS
  - 38.4.4 Power pack, Modem with SIM card and accessories for testing the remote meter reading
  - 38.4.5 Any other accessories required for observing the performance and capabilities of the meters

**39.0 QUALITY ASSURANCE PLAN:**

- 39.1 The design life of the meter shall be minimum 20 years and to prove the design life the firm shall have at least the following quality Assurance Plan.
- 39.2 The factory shall be completely dust proof.
- 39.3 The test rooms shall be temperature and humidity controlled as per relevant standards.
- 39.4 The test and calibrating equipment shall be automatic and all test equipment shall have their valid calibration certificates.
- 39.5 Meter will be tested (in case of lot test) in fully automatic test bench with ICT. No. human intervention will be allowed during test.
- 39.6 Power supplies used in test equipment shall be distortion free with sinusoidal wave forms and maintaining constant voltage, current and frequency as per the relevant standards.

**40.0 THE CHECKS TO BE CARRIED OUT DURING MANUFACTURING OF THE METERS:**

- 40.1 Meter frame dimensions tolerances shall be minimal.
- 40.2 The assembly of parts shall be done with the help of jigs and fixtures so that human errors are eliminated.
- 40.3 The meters shall be batch tested on automatic, computerized test bench and the results shall be printed directly without any human errors.

**41.0 INFORMATION TO BE FURNISHED WITH THE BID:**

The Bidder shall invariably furnish the following information along with the bid, failing which the bid shall be liable for rejection. Information shall be separately given for individual type of material offered.

- 41.1 Statement giving list of important raw materials, names of sub-suppliers for the raw materials, list of standards according to which the raw materials are tested, list of tests normally carried out on raw materials.
- 41.2 Information and copies of test certificates in respect of bought out accessories.

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- 41.3 List of manufacturing facilities available.
- 41.4 Level of automation achieved and lists of areas where manual processing exists.
- 41.5 List of areas in manufacturing process, where stage inspections are normally carried out of quality control and details of such tests and inspections.
- 41.6 List of testing equipment available with the bidder for final testing of equipment specified and test-plant limitations, if any, vis-à-vis type, special acceptance and routine tests specified in the relevant standards and this specification. These limitations shall be very clearly brought out in schedule of deviations.
- 41.7 The list of components used in the meter.
- 41.8 A detailed list of bought-out items, which are used in the manufacturing of the meter indicating the name of firms from whom these items are procured. The details of quality assurance procedures followed in respect of the bought-out items.
- 41.9 The details of testing facilities available for conducting the routine and acceptance tests and other special tests on the meter.
- 41.10 The facility available if any for conducting type test.
- 41.11 Principle of operation of the meter, outlining the methods and stages of computation of various parameters starting from input voltage and current signals including the sampling rate, if applicable.
- 41.12 The relevant documents regarding the procurement of polycarbonate material.

#### **42.0 LAB FACILITY:**

The laboratory of manufacturer must be well equipped for testing of the meters. They must have computerized standard power source and standard equipment calibrated not later than a year (or as per standard practice).

#### **43.0 DOCUMENTATION:**

- 43.1 Twenty sets of operating manuals shall be supplied to the office of the CE (DTD) for distribution at sites.
- 43.2 One set of routine test certificates shall accompany each dispatch consignment.
- 43.3 The acceptance test certificates in case pre-dispatch inspection or a routine test certificate in cases where inspection is waived shall be approved by the purchaser.

#### **44.0 REPLACEMENT OF DEFECTIVE METERS:**

The Meters declared defective by WBSEDCL within its guarantee period shall be replaced by the supplier with new ones free of cost within one month on receipt of intimation. Failure to do so within the prescribed time limit shall lead to imposition of penalty of twice the cost of meter. The same may lead to black listing even, as decided by WBSEDCL. In this connection the decision of WBSEDCL shall be final.

#### **45.0 PACKING & FORWARDING:**

- 45.1 The equipment shall be packed in cartons / crates suitable for vertical / horizontal transport as the Case may be, and suitable to withstand handling during transport and outdoor storage during transit. The supplier shall be responsible for any damage to the equipment during transit, due to improper and inadequate packing. The easily damageable material shall be carefully packed and marked with the appropriate caution symbol. Wherever necessary, proper arrangement for lifting, such as lifting hooks etc., shall be provided. Supplier without any extra cost shall supply any material found short inside the packing cases immediately.
- 45.2 The packing shall be done as per the standard practice as mentioned in IS 15707: 2006. Each package shall clearly indicate the marking details (for e.g. manufacturer's name, Sl. Nos. of meters in the package, quantity of meter, and other details as per supply order). However, the supplier shall ensure the packing is such that, the material shall not get damaged during transit.

#### **46.0 COMPONENT SPECIFICATIONS:**

The meters shall be designed and manufactured using SMT (Surface Mount Technology) components, except for power supply components, LED / LCD etc., which are PTH type. All the material and electronic power components used in the



manufacture of the meter shall be of highest quality and reputed makes so as to ensure higher reliability, longer life and sustained accuracy. The Components used for manufacture of meter shall be of high quality and the bidders shall confirm component specification as specified below in Annexure-III. Bidders shall compulsorily fill Annexure-I, Annexure-II & Annexure-III for technical qualification.

| Sl. no. | Component Function / Feature  | Requirement                                                                                                                                                                                                                                                                                                                                                      | Make / origin                                                                                                                 |
|---------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 1       | Current Element               | E-beam /spot welded CT shall be provided in the phase element and in the neutral with proper isolation.                                                                                                                                                                                                                                                          | Any make or origin conforming to IS-2705                                                                                      |
| 2       | Measurement / computing chips | The Measurement / computing chips used in the meter shall be with the Surface mount type along with the ASICs.                                                                                                                                                                                                                                                   | Analog Devices, AMS, Cyrus Logic, Atmel, SAMES, Texas Instruments, Teridian, NEC, Freescale, Renesas, Phillips                |
| 3       | Memory chips                  | The memory computing chips shall not be affected by the external parameters like sparking, high voltage spikes or electrostatic discharges.                                                                                                                                                                                                                      | National Semi Conductor, Atmel, SAMES, Texas Instruments, Teridian, ST, Microchip, Hitachi, OKI, Freescale, Renesas, Phillips |
| 4       | Display modules               | The display modules shall be well protected from the external UV radiations. The display shall be clearly visible over an angle of at least a cone of 70o. The construction of the modules shall be such that the displayed quantity shall not disturbed with the life of display. The display shall be TN type industrial grade with extended temperature range | Bonafied Technologies, Advantek , Hitachi, SONY, Hijing, Truly Semiconductor, Tianma                                          |
| 5       | Communication modules         | Communication modules shall be compatible for the RS 232 ports                                                                                                                                                                                                                                                                                                   | National Semiconductors, HP, ST, Texas Instruments, Agilent, Avago, Fairchild, Philips, Ligittek, Hitachi, Siemens, Everlight |
| 6       | Optical port                  | Optical port shall be used to transfer the meter data to meter reading instrument. The mechanical construction of the port shall be such to facilitate the data transfer easily.                                                                                                                                                                                 | National Semiconductors, Texas Instruments, HP, Agilent, Avago, Osram, Hitachi, Siemens, Philips, Everlight                   |
| 7       | Power Supply                  | The power supply shall be with the capabilities as per the relevant standards. The power supply unit of the meter shall not be affected in case the maximum voltage of the system appears to the terminals due to faults or due to wrong connections.                                                                                                            | As specified.                                                                                                                 |
| 8       | Electronic components         | The active & passive components shall be of the surface mount type & are to be handled & soldered by the state of art assembly processes.                                                                                                                                                                                                                        | National Semiconductors, Atmel, Phillips, Texas Instruments, ST, Onsemi, Hitachi, Oki, Toshiba, Freescale, Samsung.           |

|    |                        |                                                                                                                                                                                                                                        |                                                                                                                                              |
|----|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 9  | Mechanical parts       | The internal electrical components shall be of electrolytic copper & shall be protected from corrosion, rust etc. The other mechanical components shall be protected from rust, corrosion etc. by suitable plating / painting methods. | N.A.                                                                                                                                         |
| 10 | Battery                | Lithium-ion with guaranteed life of 10 years                                                                                                                                                                                           | Renata, Panasonic, Varta, , Sanyo, National, Tadiran, Sony, Duracell, Tekcell, Mitsubishi, EVE, SAFT , XENO                                  |
| 11 | RTC / Micro controller | The accuracy of RTC shall be as per relevant IEC / IS standards                                                                                                                                                                        | Dallas, Atmel, Motorola, NEC, Teridian Renesas, Texas Instruments, ST, Micro chips, Epson, Philips, NEC, OKI, Hitachi, Mitsubishi, Freescale |

#### 47.0 ANNEXURE:

The Bidder shall submit the following schedules (as per Standard Format), which is part and parcel of the Specification.

Annexure-I: Guaranteed Technical Particulars of Meter

Annexure-II: Pre-Qualification Criteria

Annexure-III: Component details

Annexure-IV: List of Documents to be submitted during sample submission

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**Annexure-I****GUARANTEED TECHNICAL PARTICULARS FOR 11KV STATIC BULK METERS**

| Sl. No. | Item Description                                           | Required Specification                                                                                                                                                                                       |
|---------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Name of manufacturer                                       | Bidder to specify                                                                                                                                                                                            |
| 2       | Type (Model No.), name, number                             | Bidder to specify                                                                                                                                                                                            |
| 3       | Standards Applicable                                       | IS 14697: 1999 read with latest amendments, IS 15707, IS 13410:1992 along with latest amendments, IS 12346:1988, CBIP Report No. 325 & its latest amendments                                                 |
| 4       | Ratings                                                    |                                                                                                                                                                                                              |
| (i)     | Accuracy class                                             | 0.5s                                                                                                                                                                                                         |
| (ii)    | Rated Voltage                                              | Ph-N : 110 / $\sqrt{3}$ V (63.5 V)                                                                                                                                                                           |
| (iii)   | Basic Current ( $I_b$ )                                    | 5A CT Ratio 50/ 5 A                                                                                                                                                                                          |
| (iv)    | Rated Frequency                                            | 50 Hz $\pm$ 5%                                                                                                                                                                                               |
| (v)     | Power Factor Range                                         | 0 Lag-unity-0 Lead                                                                                                                                                                                           |
| (vi)    | Meter Constant (imp/kWh)                                   | Bidder to specify                                                                                                                                                                                            |
| 5       | Minimum Starting Current                                   | 0.1% of $I_b$                                                                                                                                                                                                |
| 6       | Power Loss at Rated Frequency & Reference Temperature in : |                                                                                                                                                                                                              |
| (i)     | Current Circuit                                            | Less than 1.0 VA                                                                                                                                                                                             |
| (ii)    | Voltage Circuit                                            | Less than 1.5 W & 10 VA                                                                                                                                                                                      |
| 7       | Allowable Voltage Variation                                | 70% to 120% of Rated Voltage                                                                                                                                                                                 |
| 8       | Parameters displayed                                       | To be mentioned                                                                                                                                                                                              |
| 9       | Dynamic range                                              | As per IS 14697                                                                                                                                                                                              |
| 10      | MD reset Provisions                                        | Provisions to reset MD:<br>a) Using CMRI through authenticated Command<br>b) Using Remote Communication Command<br>b) Automatic Resetting at preset date & time<br>c) Manual Resetting through push button   |
| 11      | Display :                                                  |                                                                                                                                                                                                              |
| (i)     | Type of Register                                           | LCD suitable for operation up to 60°C                                                                                                                                                                        |
| (ii)    | No. of Digits of Display                                   | 7 Full Digits (minimum)                                                                                                                                                                                      |
| (iii)   | Height of character                                        | 10 X 5 mm (minimum)                                                                                                                                                                                          |
| (iv)    | Auto display mode & scroll mode                            | Both required                                                                                                                                                                                                |
| (v)     | Type of Push Button for scroll mode                        | Spring loaded push button to be provided on top cover of meter                                                                                                                                               |
| 12      | Non Volatile Memory                                        | To be provided as per IS specification                                                                                                                                                                       |
| 13      | Provision for taking reading during power off condition    | During power off condition also, meter shall display reading with backup power through the push button provided on the meter. All data downloading facility shall have to be provided in power off condition |
| 14      | Principle of operation                                     | As per applicable standards                                                                                                                                                                                  |
| 15      | MD Integration period                                      | 15 min                                                                                                                                                                                                       |
| 16      | Weight of meter                                            | Bidder to specify                                                                                                                                                                                            |
| 17      | Dimension                                                  | Bidder to specify                                                                                                                                                                                            |
| 18      | Guarantee                                                  | 5 & ½ years from the date of supply                                                                                                                                                                          |
| 19      | Outline drawings & Leaflets                                | Bidder to provide                                                                                                                                                                                            |
| 20      |                                                            |                                                                                                                                                                                                              |
| (i)     | Remote meter readout facility                              | Must have provision                                                                                                                                                                                          |
| (ii)    | Communication Protocol used                                | DLMS                                                                                                                                                                                                         |

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|       |                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (iii) | Sealing provision for Meter & Optical Port                                               | At least two fixing screws for fixing meter cover with the meter base shall be provided. Each Screw shall have two holes; one for fitting/fixing of manufacturer's seal and the other for utility seal. The arrangement shall be in such a manner that any access to the working part of the meter body will not be possible without breaking/ tampering / removing the said seals. Suitable arrangement shall be made for fitting/fixing of utility seal at two sides of meter terminal cover in such a manner that any access to the terminal cannot be possible without removing the seals. There shall also be provision for sealing the optical port. The meter cover shall be permanently fixed to the base by ultrasonic welding or by any other technic which is either equally or more efficacious so that cover cannot be opened without breaking, i.e. the meter shall be break to open type. Meter shall have an indication in its display if top cover is removed. |
| (iv)  | Baud rate of data transmission                                                           | 9600 bps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| (v)   | Required software to be resident in HHU                                                  | Bidder to provide                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| (vi)  | Ultrasonic welding ( or any other technology, equally or more efficacious ) of body      | Must be provided                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| (vii) | Manufacturer's Seal                                                                      | Must be provided                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 21    | Base Computer Software                                                                   | Compatible with Windows 7 or higher                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 22    | Type Test Certificates                                                                   | Bidders to provide                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 23    | Time of Day Zones (Selectable)                                                           | 3 Zones to be provided with provision of 8 Zones                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 24    | Whether meter measures both fundamental & harmonic energy                                | As per specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 25    | Real Time Clock Accuracy                                                                 | Maximum drift +3 min per annum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 26    | Battery for Real Time Clock                                                              | It shall be Lithium-ion / Lithium battery having at least 10 years of life                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 27    | Anti Tamper Features                                                                     | Must be as per Tamper Logic provided by WBSEDCL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 28    | Effect of Accuracy under tamper condition                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 29    | Drift in Accuracy of measurement with time                                               | No Drift with time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 30    | Name plate details                                                                       | As per relevant IS and specified Technical Specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 31    | Type of calibration                                                                      | Meter shall be software calibrated at factory & there shall not have any provision for mechanical form of calibration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 32    | Type of mounting                                                                         | Projection Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 33    | Testing facility                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| (i)   | Fully automatic computerized meter test bench with print out facility shall be available | Must be available                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| (ii)  | Make and Sl. No. of Test bench with calibration validity                                 | Bidders to specify                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| (iii) | Accuracy of ESS duly calibrated                                                          | Bidders to specify                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| (iv)  | Following in house testing facility shall be available:                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|       | (i) AC high voltage test                                                                 | Must be available                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|       | (ii) Insulation test                                                                     | Must be available                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

*[Handwritten signatures and marks are present at the bottom of the page.]*



|       |                                                                            |                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | (iii) Test of no load condition                                            | Must be available                                                                                                                                                                                                                                                                        |
|       | (iv) Test of Starting condition                                            | Must be available                                                                                                                                                                                                                                                                        |
|       | (v) Test on Limits of error                                                | Must be available                                                                                                                                                                                                                                                                        |
|       | (vi) Power loss in voltage and current circuit                             | Must be available                                                                                                                                                                                                                                                                        |
|       | (vii) Test of Repeatability of error                                       | Must be available                                                                                                                                                                                                                                                                        |
|       | (viii) Test of meter constant                                              | Must be available                                                                                                                                                                                                                                                                        |
|       | (ix) Power loss in voltage & current circuit                               | Must be available                                                                                                                                                                                                                                                                        |
|       | (x) Test of Magnetic influence                                             | Must be available                                                                                                                                                                                                                                                                        |
| 34    | Data retention by NVM without battery backup and under power off condition | 10 years                                                                                                                                                                                                                                                                                 |
| 35    | Type of material used                                                      |                                                                                                                                                                                                                                                                                          |
| (i)   | Base                                                                       | High impact strength, non-hygroscopic, fire retardant, fire resistant, UV stabilized poly carbonate (LEXAN 503R or equivalent). Meter base with LEXAN 500R/143R may be accepted subject to verification                                                                                  |
| (ii)  | Cover                                                                      | High impact strength, non-hygroscopic, fire retardant, fire resistant, UV stabilized transparent poly carbonate (LEXAN 943A or equivalent)                                                                                                                                               |
| (iii) | Terminal Block                                                             | Material High impact strength not hygroscopic, fire retardant, fire resistant, UV stabilized poly carbonate (LEXAN 503R or equivalent) Barrier of adequate size shall be provided between phase and neutral. Terminal block with LEXAN 500R/143R may be accepted subject to verification |
| (iv)  | Terminal Cover                                                             | High Impact strength, non-hygroscopic, fire retardant, fire resistant, glass reinforced poly carbonate (Transparent) & non detachable with hinging arrangement (LEXAN 943A or equivalent). Terminal cover with LEXAN 143 may be accepted subject to verification                         |
| 36    | Screw                                                                      |                                                                                                                                                                                                                                                                                          |
| (i)   | Material                                                                   | Tin/Nickel-plated Brass                                                                                                                                                                                                                                                                  |
| (ii)  | Size                                                                       | As per relevant IS Standards                                                                                                                                                                                                                                                             |
| 37    | Internal Diameter of Terminal Hole                                         | 5.5 mm (minimum)                                                                                                                                                                                                                                                                         |
| 38    | Center to Center clearances between adjacent terminals                     | 13 mm                                                                                                                                                                                                                                                                                    |
| 39    | Security Profiles                                                          |                                                                                                                                                                                                                                                                                          |
| (i)   | Basic Security                                                             | To be provided                                                                                                                                                                                                                                                                           |
| (ii)  | Advance Security                                                           | To be provided                                                                                                                                                                                                                                                                           |

**Annexure-II**  
**PRE-QUALIFICATION CONDITIONS FOR THREE PHASE STATIC METERS**

| Sl. No. | Requisite Qualification                                                                                                                                             | Remarks  |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1       | Bidder shall have valid BIS certification for the offered meter                                                                                                     | Yes / No |
| 2       | Bidder shall have ISO 9001 certification                                                                                                                            | Yes / No |
| 3       | Bidder shall be manufacturer of static meters having supplied Static 1-phase or 3-phase meters with memory and LCD display to Electricity Utilities in last 2 years | Yes / No |

Static 3 Phase 4 Wire 11 kV CT & PT Operated AMR Compatible DLMS Compliant Category C1 Tri Vector Energy Meter For Bulk Consumer

|   |                                                                                                                                                                |          |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 4 | Bidder shall have Type Test certificate from any NABL accredited laboratory for the type of offered meter not older than three years                           | Yes / No |
| 5 | Bidder shall have dust free, static protected environment for manufacture, assembly and testing of meters                                                      | Yes / No |
| 6 | Bidder shall have automatic computerized test bench for lot testing of meters                                                                                  | Yes / No |
| 7 | Bidder shall have facilities of Oven for ageing test                                                                                                           | Yes / No |
| 8 | Bidder shall have certificate from any NABL accredited laboratory for immunity of the type of meter offered, against magnetic influence of 0.2 T AC & 0.5 T DC | Yes / No |

**Annexure-III**  
**COMPONENT DETAILS**

| Sl. No. | Component Function / Feature  | As per Requirement | Make / origin |
|---------|-------------------------------|--------------------|---------------|
| 1       | Current Element               |                    |               |
| 2       | Measurement / Computing chips |                    |               |
| 3       | Memory chips                  |                    |               |
| 4       | Display modules               |                    |               |
| 5       | Communication modules         |                    |               |
| 6       | Optical port                  |                    |               |
| 7       | Power Supply                  |                    |               |
| 8       | Electronic components         |                    |               |
| 9       | Mechanical parts              |                    |               |
| 10      | Battery                       |                    |               |
| 11      | RTC / Micro controller        |                    |               |

**Annexure-IV**  
**LIST OF DOCUMENTS TO BE SUBMITTED DURING SAMPLE SUBMISSION**

| Sl. No. | DOCUMENTS TO BE SUBMITTED                                                             |
|---------|---------------------------------------------------------------------------------------|
| 1       | Attested copy of type test reports from NABL accredited laboratory                    |
| 2       | Attested copy of BIS certificates of the same type of meter submitted as sample       |
| 3       | Attested certificates as regards material used for meter case, cover & terminal block |
| 4       | Annexure - II as per tender document                                                  |
| 5       | Annexure - III as per tender document                                                 |
| 6       | Operating manual of the meter submitted                                               |

P K Banerjee  
CE: P&E-Dist

A Banerjee  
Advisor(Engg)

S K Ghosh  
CE: P&CD

S Chakraborty  
CE: DTD

A Bir  
CE: CCD

D K Pal  
CE: REM

S K Ghatak  
ACE: REM

A Mitra  
ACE: DTD

S Das  
ACE: P&CD

P K Nayek  
ACE: Project-III

M M De  
ACE: Dist

D K Pal  
ACE: Comm

S K Guha  
ACE: P&E

S N Mukherjee  
SE(E): P&E

A K Biswas  
DE(E): DTD

Static 3 Phase 4 Wire 11 kV CT & PT Operated AMR Compatible DLMS Compliant Category C1 Tri Vector Energy Meter For Bulk Consumer